

Chapter	Title	Total periods	Duration
Ch 2	Acidic, Basic, and Neutral	15	40 mins × 15

C No.	Targeted NCF competencies	Justification
C-1.1	Classifies matter based on observable physical (solid, liquid, gas, shape, volume, density, transparent, opaque, translucent, magnetic, non-magnetic, conducting, non-conducting) and chemical (pure, impure; acid, base; metal, non-metal; element, compound) characteristics	The chapter's central content domain is the classification of matter by chemical characteristics. C-1.1 explicitly lists 'acid, base' and 'metal, non-metal' as examples of chemical classification. The chapter classifies substances as acidic, basic, or neutral using indicators (litmus, red rose extract, turmeric paper, olfactory indicators), identifies chemical characteristics (sour taste, soapy feel, colour change behaviour), and lists acids present in common edible substances. This is a direct, complete match to C-1.1's chemical classification definition.
C-1.2	Describes changes in matter (physical and chemical) and uses particulate nature to represent the properties of matter and the changes	The chapter's second structural strand is the neutralisation reaction — a chemical change where acid and base react to produce salt, water, and heat. C-1.2 covers changes in matter (physical and chemical). The chapter dedicates a full section to the neutralisation reaction, its chemical equation (Acid + Base → Salt + Water + Heat), and its applications in daily life (ant bite treatment, soil correction, industrial waste treatment). This is an independently developed chemical change strand, not a vehicle for classification.

Stage No.	Progression Stage
1	Observe and sort by indicator response
2	Classify and compare across multiple indicators
3	Explain the neutralisation reaction as a chemical change
4	Apply classification and neutralisation reasoning to real-world situations

The learning progression each group moves through, from first contact to confident practice — the staged, inquiry-led sequence the NCF asks of Science.

This assessment is formative. Every question traces to a specific learning outcome, and each outcome is derived from an NCF competency this chapter develops. Rather than testing recall or asking whether a student “understood” a topic, each item asks for an observable demonstration — a response, a classification, a justification, or a product — of the thinking the lessons built. The questions are grouped by stages, and each shows the learning outcome it assesses and its cognitive demand, so you can see exactly what each question checks and why.

STAGE 1

Observe and sort by indicator response

Period 1	40 min	The Science Fair Mystery: What Are Indicators?	Pedagogy: <i>Discovery approach</i>
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Materials: Whiteboard or blackboard, Printed or narrated science fair scenario text

Teacher notes: This first unit sets the chapter's starting point by anchoring classification in a concrete mystery before any formal vocabulary is introduced. Watch for the common conflation of sour taste with all acids — students often assume only foods can be acidic; gently note that many non-food substances are also acidic or basic. A useful checkpoint: students are ready to move on when they can articulate that an indicator is something that shows a visible change, not just that it 'reacts'.

8 min	Teacher narrates the science fair story of Ashwin and Keerthi's hidden message and poses the central question: what made the paper change colour?
12 min	Students individually write down examples of sour-tasting, bitter-tasting, and soapy-feeling substances they know from daily life, grouping them by sensory property.
12 min	Class shares their lists; teacher introduces the term 'indicator' and 'acid-base indicator' as a substance that shows different colours in the presence of acids and bases, connecting it to the mystery.
8 min	Students revise their hypothesis in writing, now using the word 'indicator' to explain the turmeric paper observation.

Period 2	40 min	Litmus Test: Sorting Household Substances	<i>Hands-on Science</i>
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Materials: Blue and red litmus paper strips, Small cups or wells for samples, Lemon juice, soap solution, baking soda solution, vinegar, lime water, tap water, sugar solution, salt solution, amla juice, tamarind water, washing powder solution, Results recording sheet

Teacher notes: The previous unit established what an indicator is in principle; this unit is where students first apply that idea with physical materials. A persistent misconception is that a neutral substance 'does something' to one of the papers — remind students that no change in either paper is the positive result for neutral. If groups finish early, prompt them to notice which group has the most substances from their everyday kitchen.

6 min	Teacher demonstrates the litmus paper testing technique, emphasising that both blue and red strips must be used for each sample.
22 min	Student groups systematically test all provided samples, recording observed colour changes and provisional groupings in a two-column results table.
8 min	Groups compare their groupings across the class; discrepancies are discussed and resolved using the rule: blue turns red = acidic, red turns blue = basic, no change = neutral.
4 min	Students write one sentence summarising the rule for each group based on their own observations.

Period 3	40 min	Taste, Touch, and Tell: Sensory Properties of Acids and Bases	<i>Hands-on Science</i>
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Materials: Safe edible acidic samples: lemon juice, tamarind water, amla juice (small quantities), Dilute soap solution (for feel only), Matching worksheet (acid name to food source), Results table from previous unit

Teacher notes: Building on the litmus sorting from the previous unit, this unit adds a sensory dimension to the classification — students move from colour change evidence to direct sensory evidence. Watch for the misconception that all sour things are equally acidic or that sourness is the only property of acids; basic substances have their own distinct sensory signature. Ensure students understand they must never taste unlabelled or non-food substances — the safety boundary is clear.

8 min	Teacher recaps the litmus groupings from the previous unit and introduces Activity 2.2: students will link taste to the acidic group.
12 min	Students taste safe edible acidic samples (lemon juice, tamarind water, amla juice) and note the sour sensation; they observe the slippery feel of dilute soap solution on their fingertips.
12 min	Teacher introduces citric acid, tartaric acid, ascorbic acid, and oxalic acid with their food sources; students complete a matching task pairing acid name to common food.
8 min	Students add a 'sensory property' column to their litmus results table from the previous unit, reinforcing the link between chemical group and physical sensation.

Period 4	40 min	Red Rose Extract: A Second Natural Indicator	Hands-on Science
Materials: Fresh red rose petals, Mortar and pestle or stone for crushing, Filter paper or muslin cloth, Funnel and collection cup, Same household samples as in the litmus activity, Small dropper or pipette			
Teacher notes: Students come to this unit having already sorted substances with litmus, so this unit reinforces classification while introducing a second independent indicator. A common error is pressing the petals too lightly, yielding a faint extract that gives ambiguous colour changes — encourage a firm crush. The cross-check between two indicators is the key reasoning move here; students who see matching groupings are building confidence in classification as a reproducible operation.			
8 min	Teacher demonstrates Activity 2.3: crushing red rose petals and filtering to obtain a clear coloured extract, naming the product an 'indicator solution'.		
20 min	Student groups test the full set of samples with rose extract, recording the colour produced (red for acidic, green for basic) in a new results column.		
8 min	Groups compare their rose extract groupings with their earlier litmus groupings; they identify any disagreements and discuss whether the sample itself or the technique could explain them.		
4 min	Students list at least three other plant materials mentioned in the chapter (beetroot, purple cabbage, turmeric, jamun, hibiscus) and predict which of their tested substances would affect those indicators.		

Period 5	40 min	Turmeric Paper and Olfactory Indicators	Hands-on Science
Materials: Turmeric powder or paste, Filter paper strips, Tray for soaking and drying, Acidic, basic, and neutral samples from earlier activities, Onion pieces soaked in water (for olfactory indicator strips), Student notebooks			
Teacher notes: This unit completes the three-indicator exploration begun in units 2 and 4, and students should now be in a position to compare all three side by side. A frequent confusion is thinking turmeric paper should turn a different colour for acidic substances — explicitly address that 'no change' for both acidic and neutral is turmeric paper's designed limitation, not an experimental error. The comparative summary table at the end of this unit is a strong readiness checkpoint before the class moves into the next stage.			
10 min	Students prepare turmeric paper strips (Activity 2.5): soaking filter paper in turmeric paste, drying briefly, then cutting into strips ready for testing.		
14 min	Groups test acidic, basic, and neutral samples against turmeric paper, recording colour change for each; they discuss why turmeric paper cannot distinguish acidic from neutral.		
12 min	Students prepare and test onion-soaked strips (Activity 2.6), recording whether the odour of the strip changes for acidic or basic samples to establish olfactory indicators.		
4 min	Students summarise in their notebooks: what each of the three indicators (litmus, rose extract, turmeric) can and cannot detect, as a comparative table.		

Assessment

Q1 · Learning Outcome Students can correctly classify a given substance as acidic, basic, or neutral by applying at least two natural indicators and interpreting the resulting colour changes.	Cognitive Demand Understanding
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A student dips a strip of blue litmus paper into a sample of lemon juice. The paper turns red. What can the student correctly conclude about lemon juice?

- A. Lemon juice is acidic. ✓**
- B.** Lemon juice is basic.
- C.** Lemon juice is neutral.
- D.** Lemon juice will also turn red litmus paper red.

Answer section

Correct answer: A

What each choice reveals

- B** Choosing 'basic' indicates the student has reversed the litmus rule — confusing which paper type responds to acids versus bases.

- C** Choosing 'neutral' indicates the student equates any colour change with an inconclusive result, not recognising that blue-to-red is a definitive acidic signal.
- D** Choosing this option reveals a misconception that both litmus papers respond identically to the same substance, missing that red litmus does not change colour in acid.

Q2 · Learning Outcome Students can correctly classify a given substance as acidic, basic, or neutral by applying at least two natural indicators and interpreting the resulting colour changes.	Cognitive Demand Understanding
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A student tests four substances with litmus paper. Refer to the table provided. Which two substances belong to the same classification group?

Substance	Blue litmus	Red litmus
P	Turns red	No change
Q	No change	No change
R	No change	Turns blue
S	Turns red	No change

- A.** P and R
- B. P and S ✓**
- C.** Q and R
- D.** Q and S

Answer section

Correct answer: B

What each choice reveals

- C** Choosing Q and R shows the student is pairing substances that each caused a change in one paper but has not noticed that the changes are in opposite papers — conflating any change with the same classification.
- A** Choosing P and R indicates the student is grouping by 'causes a change' without attending to which paper changes — missing that red-to-blue signals a different group from blue-to-red.
- D** Choosing Q and S shows the student is grouping by 'involves red litmus' without reading the direction of change, or is confusing Q's neutral result with S's acidic result.

STAGE 2
Classify and compare across multiple indicators

Period 6	40 min	Which Indicator Tells the Whole Story?	Inquiry approach
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Materials: Consolidated three-indicator results table (teacher-prepared summary of group data from units 2-5), Blank decision tree worksheet or blank paper, Whiteboard for sharing keys

Teacher notes: Students move from doing individual indicator tests in previous units to reasoning about the adequacy of evidence — a significant cognitive step. The key misconception to address is that more indicators always means more confusion; help students see that each indicator adds a distinct discriminating capacity. A group that arrives at a key that correctly identifies all three groups using the fewest steps has demonstrated strong classification reasoning.

6 min	Teacher poses the inquiry question and distributes a consolidated three-indicator results table compiled from the previous units' observations.
18 min	Groups analyse the table, identify one or more substances where turmeric paper alone gives ambiguous results (acidic versus neutral both show no change), and discuss what additional test would resolve the ambiguity.

10 min	Each group constructs a classification key: a step-by-step decision tree specifying which indicator to apply first and what result leads to which classification.
6 min	Groups share their keys; teacher facilitates a brief discussion on whether any key could be wrong and what makes a classification reliable.

Period 7	40 min	Lichens, Hydrangeas, and Indicator Sources in Nature	<i>Didactic approach</i>
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Materials: Whiteboard or display surface, Hydrangea image (optional visual), Student notebooks

Teacher notes: This unit broadens classification thinking by showing that the acid-base indicator phenomenon exists beyond the classroom — it is present in living systems. A misconception to watch for is that hydrangea and litmus 'work the same way'; the hydrangea absorbs soil chemicals through its roots, which is a different mechanism from dipping paper in a solution. The connection to the rose extract experiment (both involve plant pigments changing colour) is a productive link to draw out.

12 min	Teacher explains the Holistic Lens content: lichens as symbiotic associations of fungus and alga, found in clean-air environments, and the traditional use of lichen-derived dye to make litmus.
12 min	Teacher presents the Fascinating Facts content on hydrangea: the flower changes colour depending on soil pH, demonstrating that acid-base conditions affect natural pigments — the same principle as their rose extract work.
12 min	Students individually answer two guided questions in their notebooks, linking lichen sensitivity and hydrangea colour to concepts from their indicator experiments.
4 min	Brief whole-class share: students identify one connection between the hydrangea fact and their own rose extract observations from unit 4.

Period 8	40 min	Acharya P.C. Ray and the Scientist's Toolkit	<i>Hands-on Science</i>
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Materials: Two or three unlabelled samples prepared by teacher (from the standard set used in earlier units), Available indicator materials: litmus paper strips, rose extract, turmeric paper strips, Student notebooks with classification key from unit 6

Teacher notes: This unit consolidates Stage 2 by having students apply their classification toolkit with genuine decision-making, not just following a script. The Acharya P.C. Ray connection is more than biographical — it frames the classification work as part of a broader tradition of building reliable scientific tools from local materials. Watch for groups who default to litmus alone; prompt them to justify why one indicator is sufficient for their chosen unknown.

10 min	Teacher reads or paraphrases the Know a Scientist feature on Acharya P.C. Ray; students discuss what 'scientific toolkit' means in the context of their own three-indicator work.
20 min	Groups receive two or three unlabelled samples and classify them using their chosen indicator sequence, recording both the evidence and the reasoning for their verdict.
8 min	Groups share verdicts and indicator sequences with the class; teacher highlights that different valid sequences can reach the same classification — reliability is about evidence, not procedure uniformity.
2 min	Students note one improvement they would make to their own classification key from unit 6 based on this experience.

Assessment

Q3 · Learning Outcome Students can evaluate which indicator or combination of indicators is needed to reliably distinguish all three substance groups, and justify their choice with reference to each indicator's limitations.	Cognitive Demand Analysis
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A student wants to find out whether an unknown liquid is acidic, basic, or neutral. She tests it with turmeric paper and observes no colour change. What can she correctly conclude, and what should she do next?

- A. The liquid is acidic; turmeric paper always stays yellow for acids.
- B. The liquid is basic; turmeric paper only stays unchanged for bases.
- C. **The liquid is either acidic or neutral; she should test it with litmus paper to distinguish between the two. ✓**
- D. The liquid is neutral; no further testing is needed.

Answer section

Correct answer: C

What each choice reveals

- D** Choosing this reveals the student has not recognised turmeric paper's limitation — it cannot distinguish acidic from neutral, so a no-change result is ambiguous, not definitive for neutral.
- A** Choosing this shows the student has partially grasped turmeric's limitation but incorrectly assigned the no-change result exclusively to acidic substances, rather than to both acidic and neutral.
- B** Choosing this reveals a complete reversal of turmeric paper's behaviour — the student believes no change signals basic when in fact basic substances cause the yellow-to-red change.

Q4 · Learning Outcome

Students can evaluate which indicator or combination of indicators is needed to reliably distinguish all three substance groups, and justify their choice with reference to each indicator's limitations.

Cognitive Demand
Analysis

Red rose extract turns red in acidic solutions and green in basic solutions. Turmeric paper turns red only in basic solutions. A student tests a sample and gets a red colour with rose extract and no change with turmeric paper. What is the correct classification of the sample?

- A.** The results are contradictory; classification is impossible.
- B. The sample is acidic. ✓**
- C.** The sample is basic.
- D.** The sample is neutral.

Answer section

Correct answer: B

What each choice reveals

- C** Choosing basic reveals the student has focused on the turmeric result (no change) and incorrectly inferred that no change means basic, reversing turmeric paper's rule.
- D** Choosing neutral shows the student correctly noted that turmeric paper showed no change (ruling out basic) but then failed to use the rose extract red result to distinguish acidic from neutral.
- A** Choosing this reveals the student cannot reconcile the two indicators' results because they have not grasped that both results are consistent with acidic — rose extract turns red for acid, and turmeric shows no change for both acidic and neutral.

Q5 · Learning Outcome

Students can evaluate which indicator or combination of indicators is needed to reliably distinguish all three substance groups, and justify their choice with reference to each indicator's limitations.

Cognitive Demand
Analysis

A student has only turmeric paper available. She tests three unknown substances and observes no colour change in all three cases. Explain why she cannot classify all three substances correctly using turmeric paper alone, and state what she should do to complete the classification.

Answer section

Expected elements

- Turmeric paper turns red only in the presence of basic substances; it does not change colour for either acidic or neutral substances.
- A 'no change' result means the substance is either acidic or neutral — the student cannot tell which from turmeric paper alone.
- To distinguish acidic from neutral, the student should also test with litmus paper (or red rose extract), which responds differently to acids and neutral substances.

STAGE 3

Explain the neutralisation reaction as a chemical change

Period 9	40 min	Lemon Juice Meets Lime Water: Observing Neutralisation	<i>Hands-on Science</i>
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Materials: Lemon juice, Lime water, Blue litmus solution, Dropper or pipette, Small transparent cup or beaker, Stirring rod, Observation recording sheet

Teacher notes: This unit marks the transition from classification thinking to reasoning about a chemical change — a significant conceptual shift. Students often expect a dramatic colour change at neutralisation; address the misconception that neutralisation is an instant, binary event by emphasising the gradual colour progression they observe. The prediction at the end of the unit is a readiness checkpoint: students who can predict correctly have grasped that the acid-base balance can shift in either direction.

8 min	Teacher sets up the experiment: lemon juice with blue litmus solution turns red (acidic confirmed), and introduces the question — what happens if we add a base to this acid?
20 min	Student groups add lime water drop by drop, stirring after each addition, and record the colour of the solution at each stage in a sequential observation table.
8 min	Groups identify and mark the neutralisation point; teacher leads a brief discussion on what 'neither acidic nor basic' means in terms of the litmus evidence they have just produced.
4 min	Students write one sentence prediction: if they now add more lemon juice to the neutralised solution, what will happen and why?

Period 10	40 min	Products and Equation: Naming the Neutralisation Reaction	<i>Didactic approach</i>
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Materials: Whiteboard or blackboard, Student notebooks

Teacher notes: The previous unit gave students the observational foundation; this unit provides the conceptual framing that names what they observed as a chemical change. A common confusion is between the salt produced in neutralisation and common table salt (sodium chloride) — clarify that 'salt' here is a chemical term for a class of products, not necessarily the salt used in cooking. Students who can state both evidence types (new substance formed, heat released) are ready for the application stage ahead.

10 min	Teacher recaps the observation from the previous unit and asks: 'Were new substances formed, or were the original substances just mixed?' — building toward the distinction between physical and chemical change.
12 min	Teacher presents the word equation $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water} + \text{Heat}$ and explains each product, including the release of heat as evidence that a chemical change has occurred.
12 min	Students individually write an explanation in their notebooks distinguishing the neutralisation reaction from a physical mixing process, citing at least two pieces of evidence from the chapter.
6 min	Pair-share: students exchange notebooks and check whether their partner's explanation includes formation of new substances and energy release as distinguishing features.

Period 11	40 min	Designing a Neutralisation Experiment	<i>Inquiry approach</i>
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Materials: Whiteboard for group presentations, Student notebooks, Optional: available indicator materials if groups wish to pilot their procedure

Teacher notes: This unit deepens the chemical change reasoning from the previous unit by asking students to design rather than follow a procedure. The most productive misconception to address is the idea that adding more liquid always dilutes — students need to articulate why a colour change back toward neutral is evidence of a chemical change, not just less concentrated acid. Groups that focus on observable, testable criteria in their design are demonstrating the strongest reasoning at this stage.

6 min	Teacher poses the design challenge: 'You have an acidic solution and a choice of basic substances from the chapter — design a procedure to achieve neutralisation and prove it happened.'
18 min	Groups work through their design: choosing an acid-base pair from chapter examples, selecting an indicator, specifying steps, and predicting the result at the neutralisation point.
10 min	Each group presents their criterion for distinguishing neutralisation from mere dilution; class evaluates whether each criterion is observable.
6 min	Students individually note one modification they would make to their design to make the neutralisation point easier to detect.

Assessment

Q6 · Learning Outcome

Students can explain neutralisation as a chemical change by identifying the products (salt, water, heat), representing it with the word equation, and distinguishing it from physical mixing using observable evidence.

Cognitive Demand
Understanding

A student mixes lemon juice (acidic) and lime water (basic) in the correct proportions and observes that the blue litmus solution in the mixture changes from red back toward blue. What does this colour change indicate?

- A. A neutralisation reaction has occurred and the resulting solution is neither acidic nor basic. ✓**
- B. A physical change has occurred because the two liquids were combined.
- C. The lemon juice has been diluted until it is too weak to be detected.
- D. The lime water has turned acidic because it mixed with lemon juice.

Answer section

Correct answer: A

What each choice reveals

- C** Choosing this reveals the student is interpreting neutralisation as mere dilution — a key misconception the chapter specifically addresses — rather than as a chemical change producing new substances.
- D** Choosing this shows the student has not grasped the directionality of acid-base interaction; mixing does not make the base acidic, it results in a neutral product.
- B** Choosing this reveals the student has misclassified the event as a physical change, likely because they see two liquids being combined, and has not applied the criterion of new substance formation.

Q7 · Learning Outcome

Students can explain neutralisation as a chemical change by identifying the products (salt, water, heat), representing it with the word equation, and distinguishing it from physical mixing using observable evidence.

Cognitive Demand
Understanding

Which of the following best explains why mixing an acid and a base in sufficient quantities is classified as a chemical change and not a physical change?

- A. New substances — salt, water, and heat — are produced, and these cannot be reversed simply by separating the liquids. ✓**
- B. The colour of the indicator changes, which always signals a physical change.
- C. The resulting solution has a neutral taste, which means a new substance was formed.
- D. The two liquids mix together, which is what happens in all chemical changes.

Answer section

Correct answer: A

What each choice reveals

- B** Choosing this shows the student has confused the indicator's response (an observable signal) with the criterion for a chemical change, not recognising that the indicator itself undergoes a reversible physical change in colour.
- D** Choosing this reveals the student is using 'mixing' as the criterion for chemical change, which is a direct confusion between physical mixing and chemical reaction — a misconception the chapter explicitly addresses.
- C** Choosing this shows the student is attempting to use sensory evidence (taste) as the criterion for a chemical change rather than the formation of new substances with different chemical identities.

Q8 · Learning Outcome

Students can explain neutralisation as a chemical change by identifying the products (salt, water, heat), representing it with the word equation, and distinguishing it from physical mixing using observable evidence.

Cognitive Demand
Application

Write the word equation for the neutralisation reaction and name all three products. Then give one piece of observable evidence from the lemon juice and lime water experiment that shows a chemical change — not just a physical mixing — has occurred.

Answer section

Expected elements

- Word equation: Acid + Base → Salt + Water + Heat
- All three products named: salt, water, and heat.
- One observable piece of evidence that distinguishes chemical change from physical mixing — acceptable answers include: the litmus colour changes from red progressively toward neutral (not merely lighter red, indicating a new chemical state); heat is released during the reaction; the resulting solution can no longer be classified as acidic or basic by indicator tests, indicating the original substances are no longer present.

STAGE 4
Apply classification and neutralisation reasoning to real-world situations

Period 12	40 min	Ant Bite, Soil, and Lake: Neutralisation in Daily Life	Hands-on Science
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Materials: Student notebooks, Chapter text or teacher summary of Section 2.3 scenarios, Whiteboard for scenario sharing

Teacher notes: Students are now applying both classification and chemical change reasoning together for the first time — this unit is the hinge between Stage 3 and the final application stage. A misconception to watch for is the idea that any base can correct any acid situation regardless of quantity; encourage students to think about what 'sufficient quantities' means in the real situations. The individual extension at the end reveals whether students can transfer the reasoning to a novel context, which is the core target of the final stage.

8 min	Teacher models the ant bite scenario: formic acid is acidic, baking soda is basic, applying baking soda neutralises the acid — students see the full reasoning chain.
20 min	Groups analyse the soil correction and industrial waste situations independently, applying the same reasoning chain and writing justifications in their notebooks.
8 min	Groups share their analysis of one scenario; class checks whether the acid-base classification and corrective substance selection are consistent with chapter content.
4 min	Students individually propose one additional daily-life situation (not from the chapter) where neutralisation could be applied, and identify what the acidic or basic problem substance would be.

Period 13	40 min	Cracking the Science Fair Mystery: Full Explanation	Inquiry approach
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Materials: Student notebooks, Optional: a physical demonstration of turmeric paper and soap solution to re-examine the mystery

Teacher notes: This unit brings the chapter's narrative arc full circle, using the mystery as an application vehicle for the complete classification toolkit. A misconception to watch for is students claiming the soap 'reacts with' turmeric rather than the indicator responding to the basic nature of the soap — the correct framing is that turmeric paper detects bases, and soap is basic. Students who can write a complete, evidence-based explanation without prompting are demonstrating the application-level operation the chapter targets.

6 min	Teacher re-presents the science fair mystery from unit 1 and challenges students to now explain it fully, noting how much they now know that they did not in the first unit.
16 min	Students individually draft their full explanation in their notebooks, drawing on indicator properties, classification rules, and the chapter's turmeric paper results.
10 min	Pairs exchange explanations and identify any missing element — incorrect classification, wrong indicator rule, or unexplained colour change — providing written feedback.
8 min	Students revise their explanations based on feedback; teacher asks one or two students to share a strong explanation with the class.

Period 14	40 min	Novel Scenarios: Classifying and Correcting Unfamiliar Situations	<i>Inquiry approach</i>
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Materials: Novel scenario cards prepared by teacher (scenarios not directly named in Section 2.3 but resolvable using chapter principles), Student notebooks, Response sheets

Teacher notes: This unit is the final teaching unit before the synthesis; it should push students to apply classification and neutralisation reasoning to contexts that are not directly named in the chapter. Watch for groups that default to recalling named examples from the chapter rather than reasoning from indicator rules — the goal is transfer, not recall. The closing reflection question is diagnostic: students who describe needing multiple indicators or both classification and chemical-change evidence are at the strongest level of the chapter's target operation.

6 min Teacher distributes scenario cards and explains the task: use only what is in the chapter to classify the substance and, where relevant, suggest a corrective action.

22 min Groups work through their assigned scenarios, writing classification justifications and corrective action justifications on their response sheets.

8 min Groups present their reasoning to the class; teacher facilitates peer evaluation of whether each classification and corrective action is grounded in chapter evidence.

4 min Students write a one-sentence reflection: what would make them more confident in classifying an unknown substance they have never tested before?

Period 15	40 min	The Pickle Factory Problem: Reading a Substance Trail	<i>Collaborative Pair Analysis</i>
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Materials: Data card, one per pair (see visual aid: 'Pickle Factory Data Card'), Structured response sheet, one per student (see visual aid: 'Pickle Factory Response Sheet'), Blackboard and chalk

Teacher notes: This unit asks students to apply both chapter strands — indicator-based classification and neutralisation as a chemical change — to a continuous real-world scenario, reading a four-sample factory trail and recommending corrections. Circulate during pair work listening for two specific errors: treating 'turmeric paper unchanged' as evidence of acidity rather than neutrality-or-acidity, and choosing a corrective without checking whether the product would itself be harmful (see material). The closing share should surface both reasoning moves before you name the chapter's two strands as the tools that made the trail readable at all.

Prepared table · *Pickle Factory Data Card*

A factory inspector visited a small pickle factory and tested four liquid samples using indicator papers. The inspector recorded only the colour changes observed

Observation	Sample	Result recorded
1	Lemon brine from the pickling vat	Turns blue litmus paper red
2	Equipment rinse solution	Leaves turmeric paper unchanged; turns red litmus paper blue
3	Factory drainage water	Produces no colour change in either litmus paper
4	Stream water downstream of the factory outlet	Turns red rose extract from red to green

Prepared text · *Data card — note*

No other information was recorded.

Prepared table · *Pickle Factory Response Sheet*

Sample	Indicator result given	Nature of sample (acidic / basic / neutral)	Treat before release? (Yes / No)	Corrective substance if needed
1 — Lemon brine	Blue litmus turns red			
2 — Equipment rinse solution	Turmeric unchanged; red litmus turns blue			
3 — Factory drainage water	No colour change in either litmus			

4 — Stream water (downstream)	Red rose extract turns red to green			
Write the neutralisation reaction for one correction:				

5 min	Teacher distributes data cards face-down. Without distributing the response sheets yet, teacher reads the factory scenario aloud: a pickle factory produces four kinds of liquid — pickling brine, equipment rinse water, general drainage, and it discharges into a stream. A factory inspector tested each one with indicators and recorded only the colour changes. Teacher asks the class: 'What would you need to know, from indicator results alone, to decide whether any of these liquids is safe to release into the stream?' Take two or three spoken responses to orient the class to the task, then distribute response sheets and ask pairs to begin.
20 min	Pairs work through the data card together using the response sheet. They classify each sample (acidic, basic, or neutral) from the indicator evidence, decide which samples require treatment before release, choose a corrective substance for each problematic sample, and write one neutralisation reaction for a correction of their choice. Teacher circulates, asking pairs to justify their classification reasoning aloud rather than correcting errors directly.
12 min	Whole-class share. Teacher calls on four different pairs, one for each sample. Each pair states their classification and, if applicable, their corrective choice and justification. After each pair speaks, the class is invited to agree, add a qualification, or challenge — teacher writes the agreed classification and corrective on the board for that sample. When all four are on the board, teacher asks one pair to read their neutralisation equation aloud and invites another pair to check it by naming the acid, the base, and the products.
3 min	Teacher closes by pointing to the four completed rows on the board and naming what made it possible to read the factory's substance trail: that natural substances can serve as indicators, that different indicators reveal different things and have individual limitations, and that when an acid and a base meet, a neutralisation reaction produces salt, water, and heat — a predictable, useful chemical change. These are the chapter's two strands, and the factory trail needed both.

Assessment

Q9 · Learning Outcome

Students can classify a substance using indicator evidence and explain how neutralisation applies as a corrective chemical change in a given everyday situation, integrating both the classification and chemical change operations simultaneously.

Cognitive Demand

Application

A farmer notices that the soil on her farm turns blue litmus paper red and that her crops are not growing well. She wants to correct the soil condition. Which substance should she add, and what reaction will take place?

- A. She should add compost because the soil is too basic; compost releases acids that neutralise the base.
- B. She should add lime (a base) to the soil; the acid in the soil and the lime will undergo a neutralisation reaction producing salt, water, and heat. ✓**
- C. She should add lime water; the neutral water in lime water dilutes the acid until the soil is no longer acidic.
- D. She should add more fertiliser (which is acidic) to make the soil even more acidic so that it reverts to normal.

Answer section

Correct answer: B

What each choice reveals

- D** Choosing this reveals the student has not connected the litmus result (acid present) to the need for a base as the corrective, and instead retrieves the association between fertilisers and soil acidity without applying the classification-correction logic.
- A** Choosing this shows the student has correctly recalled the compost-for-basic-soil remedy from the chapter but has misclassified the soil as basic when the blue-litmus-turns-red result clearly identifies it as acidic.
- C** Choosing this reveals the student is still applying the dilution misconception — treating neutralisation as a physical dilution of the acid rather than a chemical reaction requiring a base.

Q10 · Learning Outcome

Students can classify a substance using indicator evidence and explain how neutralisation applies as a corrective chemical change in a given everyday situation, integrating both the classification and chemical change operations simultaneously.

Cognitive Demand

Application

A red ant bite causes pain because formic acid is injected into the skin. A student suggests applying baking soda solution to the bite. Using the word equation for neutralisation, explain why this relieves the pain. Which option correctly completes this reasoning?

- A. Baking soda is acidic; it reacts with the formic acid to produce a stronger acid that breaks down the injected acid faster.
- B. Baking soda is basic; it absorbs the formic acid into its surface without any chemical reaction, so the acid is removed physically.
- C. **Baking soda is basic; it reacts with the formic acid in the skin to produce salt, water, and heat, neutralising the acid and removing the source of pain. ✓**
- D. Baking soda is neutral; it dilutes the formic acid in the skin until the concentration is too low to cause pain.

Answer section

Correct answer: C

What each choice reveals

- D Choosing this shows the student has correctly identified that baking soda helps but has reverted to the dilution model rather than the chemical change model, misclassifying baking soda as neutral.
- A Choosing this reveals a fundamental misclassification of baking soda as acidic — a direct error about the chemical nature of a substance the chapter explicitly identifies as basic — combined with an invented (and implausible) mechanism.
- B Choosing this shows the student correctly classifies baking soda as basic but then fails to apply the chemical change model, describing a physical absorption instead of a neutralisation reaction.

Q11 · Learning Outcome

Students can classify a substance using indicator evidence and explain how neutralisation applies as a corrective chemical change in a given everyday situation, integrating both the classification and chemical change operations simultaneously.

Cognitive Demand Analysis

A factory releases waste water into a nearby lake. Fish begin to die. A scientist tests the lake water and finds that it turns blue litmus paper red. The factory owner suggests the problem will go away if the factory dilutes its waste with more water before releasing it. A local chemist disagrees and recommends adding a basic substance to the waste before release. Write a response that:

- (a) classifies the lake water using the litmus evidence;
- (b) explains, using the neutralisation reaction, why the chemist's recommendation is correct; and
- (c) explains why the factory owner's suggestion will not solve the problem.

Answer section

Look for

- Correctly classifies lake water as acidic based on the blue litmus turning red result.
- Identifies that a basic substance must be added because neutralisation requires an acid and a base — not an acid and water.
- Uses the word equation (Acid + Base → Salt + Water + Heat) or its equivalent in words to explain what happens when the basic substance is added.
- Explains that dilution with water does not produce the chemical change needed — the acidic substances remain present, just less concentrated, and do not become chemically different.
- Connects the correction to the welfare of the fish by noting that the resulting neutral solution (or less harmful solution after neutralisation) no longer poses the same chemical threat.

Q12 · Learning Outcome

Students can classify a substance using indicator evidence and explain how neutralisation applies as a corrective chemical change in a given everyday situation, integrating both the classification and chemical change operations simultaneously.

Cognitive Demand Evaluation

Three situations describe a substance causing harm because of its acid-base nature. Refer to the table provided. For each situation:

- (1) state whether the problem substance is acidic or basic based on the litmus result given;
- (2) select a corrective substance from the chapter (baking soda, lime, compost, or another named substance) and state whether it is acidic or basic;

- (3) complete the expected products column using the word equation for the neutralisation reaction. After completing the table, write two sentences explaining why selecting the wrong type of corrective substance (e.g. an acid when the problem is acidic) would fail to solve the situation.

Scenario	Litmus result of problem substance	Nature of problem substance	Corrective substance selected	Nature of corrective substance	Expected products of reaction
Bee sting on skin (bee venom is acidic)	Blue litmus turns red				
Factory waste makes stream water turn blue litmus red	Blue litmus turns red				
Soil in a field turns red litmus paper blue, reducing crop yield	Red litmus turns blue				

What to produce

- Completed table with all six empty cells filled in
- Two-sentence written explanation of why a wrong corrective substance fails

Scaffold

Use the litmus result in the table to determine the nature of each problem substance. For the corrective substance, remember: if the problem substance is acidic, the corrective must be basic, and vice versa. The expected products of any neutralisation reaction are the same regardless of which specific acid and base are involved.

Answer section

Reading this task

Format: Classification or sorting task with justification — The Final Stage implied LO requires students to simultaneously classify a substance by indicator evidence and explain neutralisation as a corrective chemical change; a classification-with-justification table naturally structures both operations together across multiple scenarios, making the integrated reasoning directly visible.

What this demonstrates: A complete response demonstrates that the student can read indicator evidence to classify a substance, select the chemically appropriate corrective substance (opposite acid-base nature), apply the neutralisation word equation to predict products, and reason about what happens when the corrective is wrong — all integrating the classification and chemical change operations from the chapter's two central competencies.

Reading the scaffold: The table structure separates the classification step from the correction step, making each reasoning move independently readable; a teacher can see exactly where a student's reasoning breaks down — at classification, at corrective selection, or at product prediction — without the error propagating invisibly into a prose response. The two-sentence explanation at the end is the integration test: it requires students to reason about the mechanism, not just recall the correct answer.

Strong vs weak markers: Strong responses: correct classification in all three rows using the stated litmus result; corrective substance matches the chemical nature required (base for acid, acid for base); products column consistently shows salt, water, and heat; the two sentences explain the failure in terms of no chemical change occurring (not merely 'it would not work'). Weak responses: classification errors despite a direct litmus clue in the table; corrective substance is the same nature as the problem substance; products column left blank or filled with 'no change'; the two-sentence explanation restates the problem without invoking the mechanism of neutralisation.